

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA5112

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

I Bhavya PS(192524017) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Annexure A

Analytical Problem Solving

1. RSA Key Generation and Security Analysis

Given:

$p = 17, q = 19$

Compute n and $\phi(n)$

Choose $e = 5$ and calculate the private key d

Encrypt the message $M = 7$

Decrypt the ciphertext and verify correctness

Analyze the impact of choosing a small prime number on RSA security

2. ElGamal Encryption Analysis

Given:

$p = 23, g = 5, \text{ private key } x = 7, \text{ random } k = 3, \text{ message } M = 10$

Compute the public key

Perform encryption to generate ciphertext pair $(C1, C2)$

Decrypt the ciphertext to retrieve the original message

Analyze how randomness (k) improves security in ElGamal

3. RSA Digital Signature Verification

Given:

$p = 11, q = 17, e = 7, \text{ message } M = 8$

Compute n and $\phi(n)$

Find private key d

Generate the digital signature for the message

Verify the signature using the public key

Analyze how digital signatures ensure non-repudiation

4. Diffie-Hellman with Attack Resistance Evaluation

Given:

$p = 29, g = 2$

User A private key = 5
User B private key = 12

Compute public keys for both users
Compute the shared secret key
Analyze what happens if an attacker intercepts and replaces public keys
Suggest a cryptographic solution to ensure authenticity

5. Hybrid Encryption System Analysis

Given:

A system uses RSA for key exchange and AES for data encryption
RSA encryption time = 5 ms per key exchange
AES encryption time = 0.5 ms per data block
Total data blocks = 500

Calculate total time taken for encryption
Compare with using only RSA for encrypting all data
Analyze why hybrid encryption is more efficient
Justify its use in modern secure communication systems

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
----------	-----------	------------------------	-----------------------------	-----------------------------	------------------------

Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. RSA Key Generation and Security Analysis

Given

- $p = 17$
- $q = 19$
- $e = 5$
- Message $M = 7$

Step 1: Compute n

$$\begin{aligned}n &= p \times q \\n &= 17 \times 19 = 323\end{aligned}$$

Answer:

$$\boxed{n = 323}$$

Step 2: Compute Euler's Totient

$$\begin{aligned}\phi(n) &= (p - 1)(q - 1) \\&= (17 - 1)(19 - 1) \\&= 16 \times 18 = 288\end{aligned}$$

Answer

$$\boxed{\phi(n) = 288}$$

Step 3: Find the Private Key d

We need

$$d \times e \equiv 1 \pmod{288}$$

Since

$$5 \times 173 = 865$$

and

$$865 = 288 \times 3 + 1$$

Therefore,

$$\boxed{d = 173}$$

Step 4: Encrypt the Message

Encryption formula:

$$C = M^e \pmod n$$
$$C = 7^5 \pmod{323}$$

Calculate:

$$7^2 = 49$$
$$7^4 = 49^2 = 2401$$
$$2401 \pmod{323} = 140$$
$$7^5 = 140 \times 7 = 980$$
$$980 \pmod{323} = 11$$

Ciphertext:

$$\boxed{C = 11}$$

Step 5: Decrypt

$$M = C^d \pmod n$$
$$M = 11^{173} \pmod{323}$$

Using modular exponentiation,

$$\boxed{M = 7}$$

Original message is recovered successfully.

Security Analysis

Using very small primes (17 and 19):

- Easy to factor $323 = 17 \times 19$
- Attacker can quickly compute $\phi(n)$
- Private key can be recovered easily
- Modern RSA uses **2048-bit or larger primes**, making factorization computationally infeasible.

2. ElGamal Encryption Analysis

Given:

- Prime number: $p = 23$
- Generator: $g = 5$
- Private key: $x = 7$
- Random number: $k = 3$
- Message: $M = 10$

Step 1: Compute the Public Key

The public key is calculated using:

$$y = g^x p$$

Substitute the given values:

$$y = 5^7 23$$

Calculation:

$$\begin{aligned} 5^2 &= 25 \equiv 2 \pmod{23} \\ 5^4 &= (5^2)^2 = 2^2 = 4 \\ 5^7 &= 5^4 \times 5^2 \times 5 \\ &= 4 \times 2 \times 5 = 40 \\ 40 23 &= 17 \end{aligned}$$

Therefore,

$$\boxed{y = 17}$$

Public Key = (23, 5, 17)

Step 2: Perform Encryption

The ciphertext consists of two values:

$$(C_1, C_2)$$

Calculate C_1

Formula:

$$C_1 = g^k p$$

Substitute the values:

$$\begin{aligned}C_1 &= 5^3 \cdot 23 \\&= 125 \cdot 23\end{aligned}$$

Since,

$$\begin{aligned}23 \times 5 &= 115 \\125 - 115 &= 10\end{aligned}$$

Therefore,

$$\boxed{C_1 = 10}$$

Calculate C_2

Formula:

$$C_2 = M \times y^k \cdot p$$

First calculate

$$17^3 \cdot 23$$

Find 17^2

$$\begin{aligned}17^2 &= 289 \\289 \cdot 23 &= 13\end{aligned}$$

Now,

$$\begin{aligned}17^3 &= 17^2 \times 17 \\&= 13 \times 17 = 221 \\221 \cdot 23 &= 14\end{aligned}$$

Now calculate

$$\begin{aligned}C_2 &= 10 \times 14 \\&= 140 \\140 \cdot 23 &= 2\end{aligned}$$

Therefore,

$$\boxed{C_2 = 2}$$

Hence the ciphertext is

$$\boxed{(10, 2)}$$

Step 3: Decrypt the Ciphertext

First compute

$$\begin{aligned} S &= C_1^x p \\ &= 10^7 23 \end{aligned}$$

Using modular exponentiation,

$$\boxed{S = 14}$$

Now find the modular inverse of 14 modulo 23.

We know,

$$\begin{aligned} 14 \times 5 &= 70 \\ 70 23 &= 1 \end{aligned}$$

Therefore,

$$14^{-1} = 5$$

Now recover the message:

$$\begin{aligned} M &= C_2 \times 14^{-1} 23 \\ &= 2 \times 5 \\ &= 10 \\ 10 23 &= 10 \end{aligned}$$

Therefore,

$$\boxed{M = 10}$$

The original message has been successfully recovered.

Analysis: How Randomness (k) Improves Security

- Every encryption uses a different random value k .
- Even if the same message is encrypted multiple times, the ciphertext will be different.
- This prevents attackers from identifying repeated messages.
- It makes ElGamal resistant to replay attacks and ciphertext analysis.
- Therefore, randomness greatly improves the security of the ElGamal cryptosystem.

3. RSA Digital Signature Verification

Given:

- $p = 11$
- $q = 17$
- $e = 7$
- Message $M = 8$

Step 1: Compute n

Formula:

$$\begin{aligned}n &= p \times q \\&= 11 \times 17 \\&\boxed{n = 187}\end{aligned}$$

Step 2: Compute Euler's Totient

Formula:

$$\begin{aligned}\phi(n) &= (p - 1)(q - 1) \\&= (11 - 1)(17 - 1) \\&= 10 \times 16 \\&\boxed{\phi(n) = 160}\end{aligned}$$

Step 3: Find the Private Key

We need

$$d \times 7 \equiv 1 \pmod{160}$$

Testing multiples:

$$\begin{aligned}7 \times 23 &= 161 \\161 \bmod 160 &= 1\end{aligned}$$

Therefore,

$$\boxed{d = 23}$$

Private key is

$$\boxed{(23, 187)}$$

Step 4: Generate the Digital Signature

Formula:

$$S = M^d n$$

Substitute the values:

$$S = 8^{23} 187$$

Using modular exponentiation,

$$\boxed{S = 173}$$

Hence,

Digital Signature = 173

Step 5: Verify the Signature

The receiver computes

$$\begin{aligned} M &= S^e n \\ &= 173^7 187 \end{aligned}$$

Using modular exponentiation,

$$= 8$$

The recovered message is

$$\boxed{8}$$

Since the recovered message equals the original message,

Signature Verified Successfully

Analysis: How Digital Signatures Ensure Non-Repudiation

- The sender signs the message using the private key.
- Only the owner of the private key can generate the signature.
- Anyone with the public key can verify it.
- If the signature is valid, the sender cannot deny sending the message.
- Thus, digital signatures provide authentication, integrity, and non-repudiation.

4. Diffie–Hellman with Attack Resistance Evaluation

Given:

- Prime number $p = 29$
- Generator $g = 2$
- User A private key = 5
- User B private key = 12

Step 1: Compute User A's Public Key

Formula:

$$\begin{aligned} A &= g^a p \\ A &= 2^5 29 \\ 2^5 &= 32 \\ 32 29 &= 3 \end{aligned}$$

Therefore,

$$\boxed{A = 3}$$

Step 2: Compute User B's Public Key

Formula:

$$\begin{aligned} B &= g^b p \\ &= 2^{12} 29 \\ 2^{12} &= 4096 \\ 4096 29 &= 7 \end{aligned}$$

Therefore,

$$\boxed{B = 7}$$

Step 3: Compute Shared Secret Key

User A computes

$$\begin{aligned} K &= B^a 29 \\ &= 7^5 29 \end{aligned}$$

Using modular arithmetic,

$$\boxed{K = 16}$$

User B computes

$$\begin{aligned} K &= A^b \text{ } 29 \\ &= 3^{12} \text{ } 29 \end{aligned}$$

Using modular arithmetic,

$$\boxed{K = 16}$$

Hence,

$$\boxed{\text{Shared Secret Key} = 16}$$

Analysis: If an Attacker Replaces Public Keys

If an attacker intercepts the communication:

- The attacker replaces User A's public key with their own.
- The attacker also replaces User B's public key with another key.
- User A unknowingly shares a key with the attacker.
- User B also unknowingly shares a different key with the attacker.
- The attacker can decrypt, modify, and re-encrypt every message before forwarding it.

This is called a **Man-in-the-Middle (MITM) Attack**.

Cryptographic Solution

To prevent this attack:

- Use Digital Signatures.
- Use Digital Certificates.
- Use Public Key Infrastructure (PKI).
- Use authenticated Diffie–Hellman protocols such as TLS.

These methods verify that the exchanged public keys belong to the intended users.

5. Hybrid Encryption System Analysis

Given:

- RSA key exchange time = **5 ms**
- AES encryption time = **0.5 ms per block**
- Number of data blocks = **500**

Step 1: Calculate AES Encryption Time

Formula:

$$\begin{aligned}\text{AES Time} &= \text{Number of Blocks} \times \text{Time per Block} \\ &= 500 \times 0.5 \\ &= 250 \text{ ms}\end{aligned}$$

Step 2: Calculate Total Hybrid Encryption Time

Formula:

$$\begin{aligned}\text{Total Time} &= \text{RSA Time} + \text{AES Time} \\ &= 5 + 250 \\ &= 255 \text{ ms}\end{aligned}$$

Therefore,

$\text{Total Hybrid Encryption Time} = 255 \text{ ms}$
--

Step 3: Using Only RSA

RSA encrypts every data block.

Formula:

$$\begin{aligned}500 \times 5 \\ = 2500 \text{ ms}\end{aligned}$$

Therefore,

$\text{RSA Only Time} = 2500 \text{ ms}$
--

Step 4: Compare the Two Methods

Method	Total Time
Hybrid Encryption	255 ms
RSA Only	2500 ms

Hybrid encryption is approximately **10 times faster** than using RSA alone.

Analysis: Why Hybrid Encryption is More Efficient

- RSA is computationally expensive and slower for encrypting large amounts of data.
- AES is a symmetric encryption algorithm that is much faster and suitable for bulk data encryption.
- Hybrid encryption uses RSA only once to securely exchange the AES key, while AES encrypts all the data efficiently.
- This combination reduces processing time and improves overall system performance.